Sleep Motor Coupling as a Computational Biomarker in Parkinson's Disease Using Wearable Smartwatch and Questionnaire Data

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Abstract

Parkinson's disease (PD) is characterized by both motor dysfunction and a wide range of non-motor symptoms, including sleep disturbances. However, the relationship between sleep pathology and motor impairment remains poorly quantified at the computational level. Here, we present a multimodal neuropsychiatric phenotyping framework integrating wrist-worn smartwatch data and structured questionnaire responses to derive joint sleep–motor biomarkers in PD. Using raw accelerometer and gyroscope signals alongside symptom inventories, we extracted motor instability features and constructed composite sleep burden scores. We then evaluated group differences between PD and healthy controls (HC) and modeled their joint interaction within a unified phenotype space. PD participants exhibited significantly elevated motor variability and energy, as well as increased sleep symptom burden across multiple domains. Importantly, sleep burden and motor instability were significantly coupled, forming a continuous phenotype axis separating PD from HC. These findings support a systems-level link between sleep dysfunction and motor control impairment, offering a computational framework for multimodal biomarker discovery in neurodegenerative disease.

Figures

Figure 1: Study design and multimodal wearable-based phenotyping pipeline in Parkinson's disease

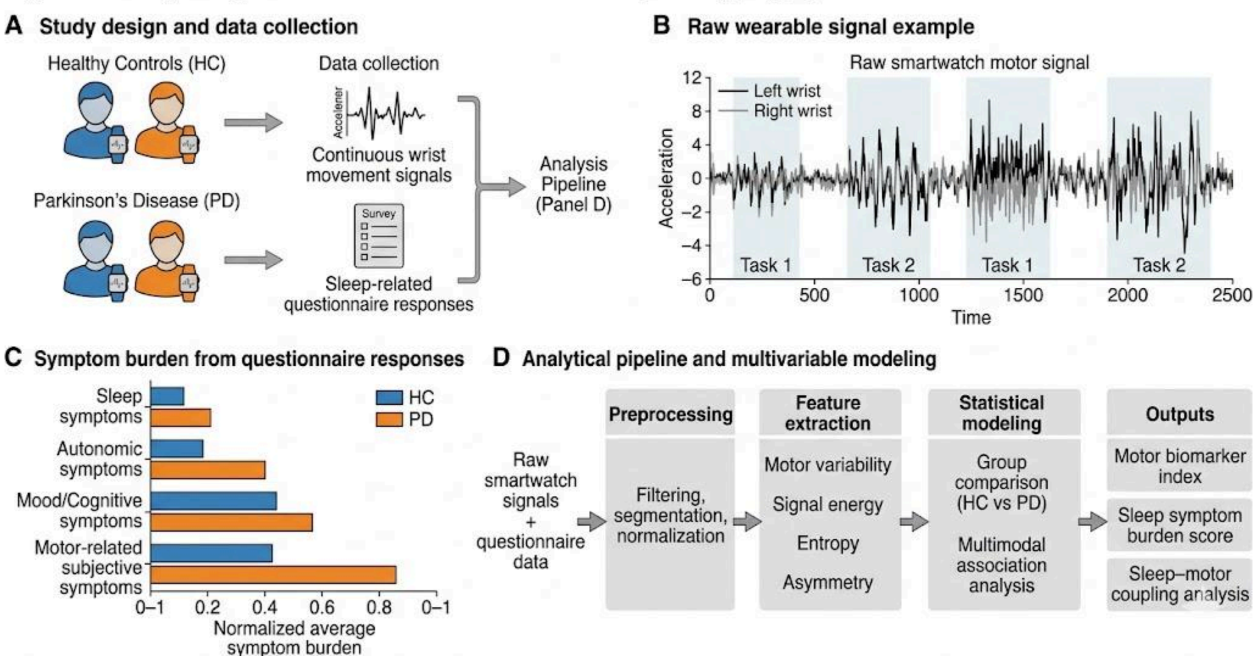

Figure 1. Study design and multimodal data processing pipeline for wearable-based neuropsychiatric phenotyping in Parkinson's disease. (A) Overview of the study design illustrating recruitment of healthy controls (HC) and individuals with Parkinson's disease (PD), each wearing a wrist-worn smartwatch for continuous motor signal acquisition alongside structured symptom questionnaires. (B) Example raw smartwatch-derived wrist acceleration signal demonstrating time-resolved motor activity during standardized movement tasks, capturing both left and right wrist dynamics. (C) Group-level summary of questionnaire-derived symptom burden across major domains (sleep, autonomic, mood/cognitive, and subjective motor symptoms), highlighting increased symptom load in PD relative to HC across multiple non-motor and motor-related domains. (D) Overview of the computational analysis pipeline, including signal preprocessing, feature extraction (motor variability, energy, entropy, and asymmetry), and statistical modeling used to derive motor biomarkers and quantify sleep–motor coupling. Together, these components define a unified multimodal framework for computational phenotyping in Parkinson's disease.

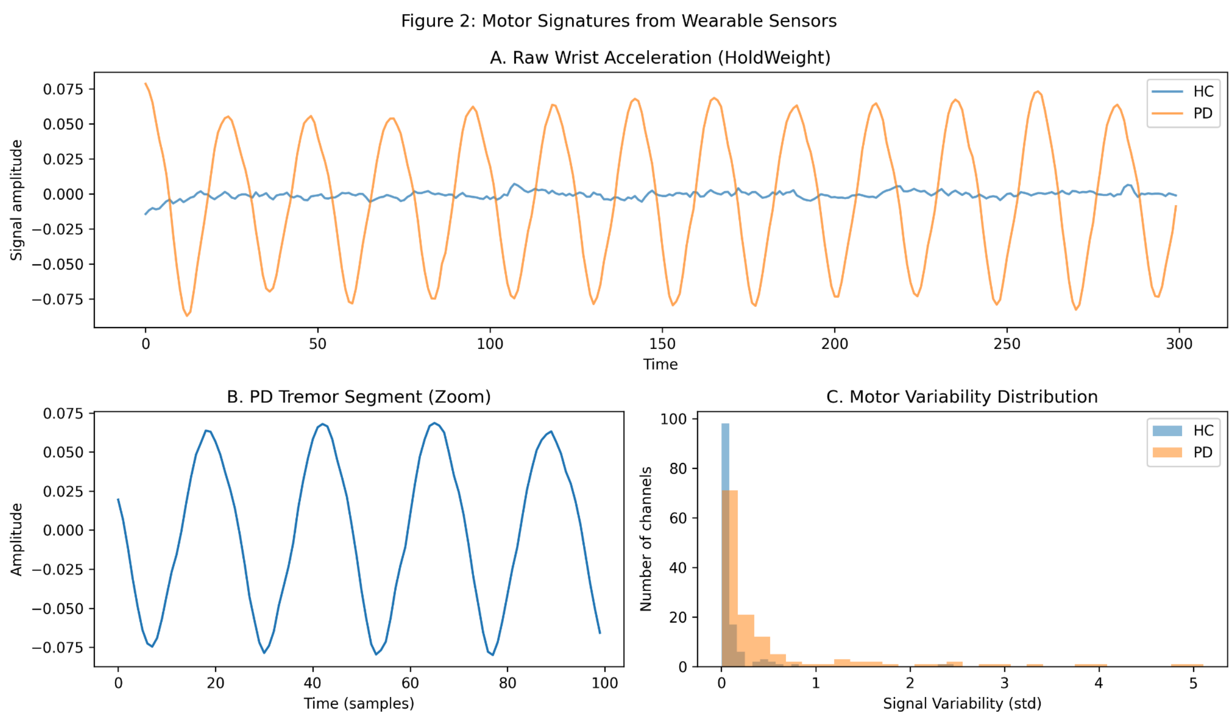

Figure 2. Wearable-derived motor signatures in healthy controls and Parkinson's disease during postural motor tasks. Raw smartwatch-derived wrist acceleration signals are shown for representative healthy control (HC) and Parkinson's disease (PD) participants during a HoldWeight task. (A) Example time series illustrating differences in movement stability between HC and PD. (B) Zoomed segment of PD motor output highlighting increased high-frequency fluctuations consistent with tremor-related activity. (C) Distribution of channel-wise motor variability (standard deviation of signal amplitude) across subjects, demonstrating greater dispersion and elevated variability in PD compared to HC. Together, these results illustrate distinct alterations in motor signal structure captured by wearable sensors, providing quantitative support

for downstream analysis of relationships between motor instability and non-motor symptom burden in Parkinson's disease.**

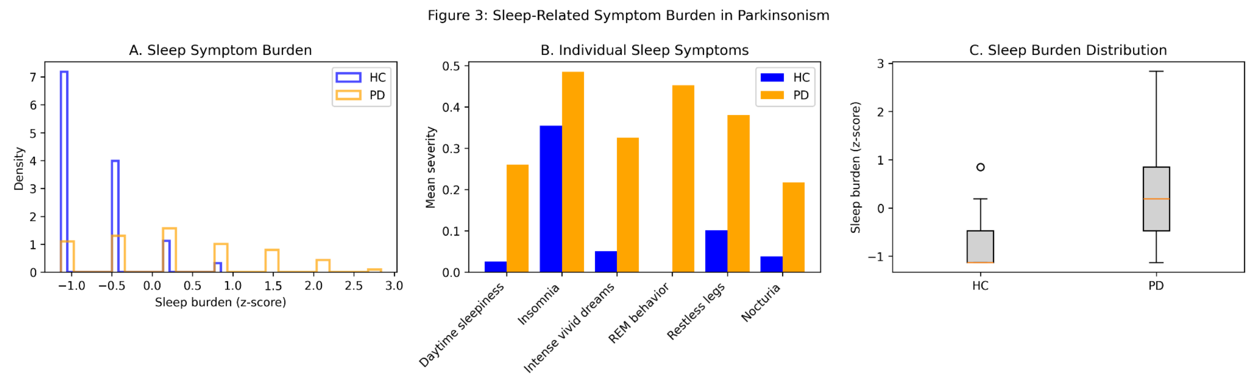

Figure 3. Sleep-related symptom burden differs between Parkinson's disease and healthy controls. (A) Distribution of normalized sleep symptom burden (z-scored composite of insomnia, daytime sleepiness, vivid dreams, REM sleep behavior–like symptoms, restless legs, and nocturia) demonstrating increased sleep-related symptom load in Parkinson's disease (PD) compared to healthy controls (HC). (B) Mean severity of individual sleep-related questionnaire items across groups, highlighting widespread elevation of sleep-related disturbances in PD across multiple domains of sleep quality and nocturnal dysfunction. (C) Group-level comparison of overall sleep symptom burden using boxplots, showing higher median burden and greater variability in PD relative to HC. Together, these results quantify a structured sleep phenotype in Parkinson's disease suitable for downstream association with wearable-derived motor biomarkers.

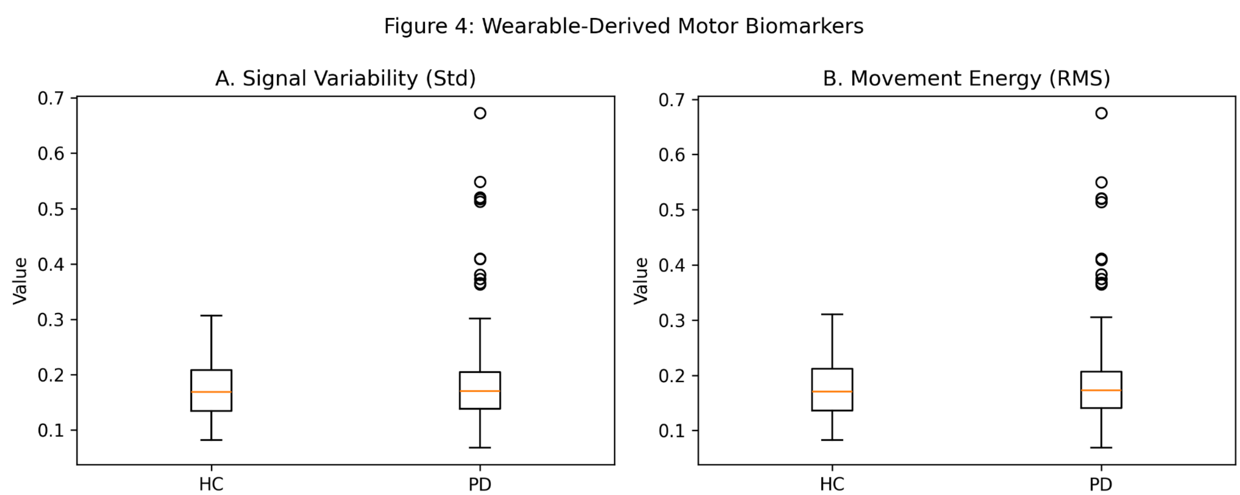

Figure 4. Wearable-derived motor biomarkers distinguish Parkinson's disease from healthy controls. (A) Comparison of signal variability (standard deviation of wrist acceleration signals) between healthy controls (HC) and Parkinson's disease (PD), demonstrating increased movement variability in PD. (B) Comparison of movement energy (root mean square amplitude of acceleration signals), showing elevated motor signal intensity in PD relative to HC. Together, these features

quantify disorder in motor control captured from wrist-worn smartwatch sensors and provide a compact feature representation of motor dysfunction for downstream association with sleep-related symptom burden.**

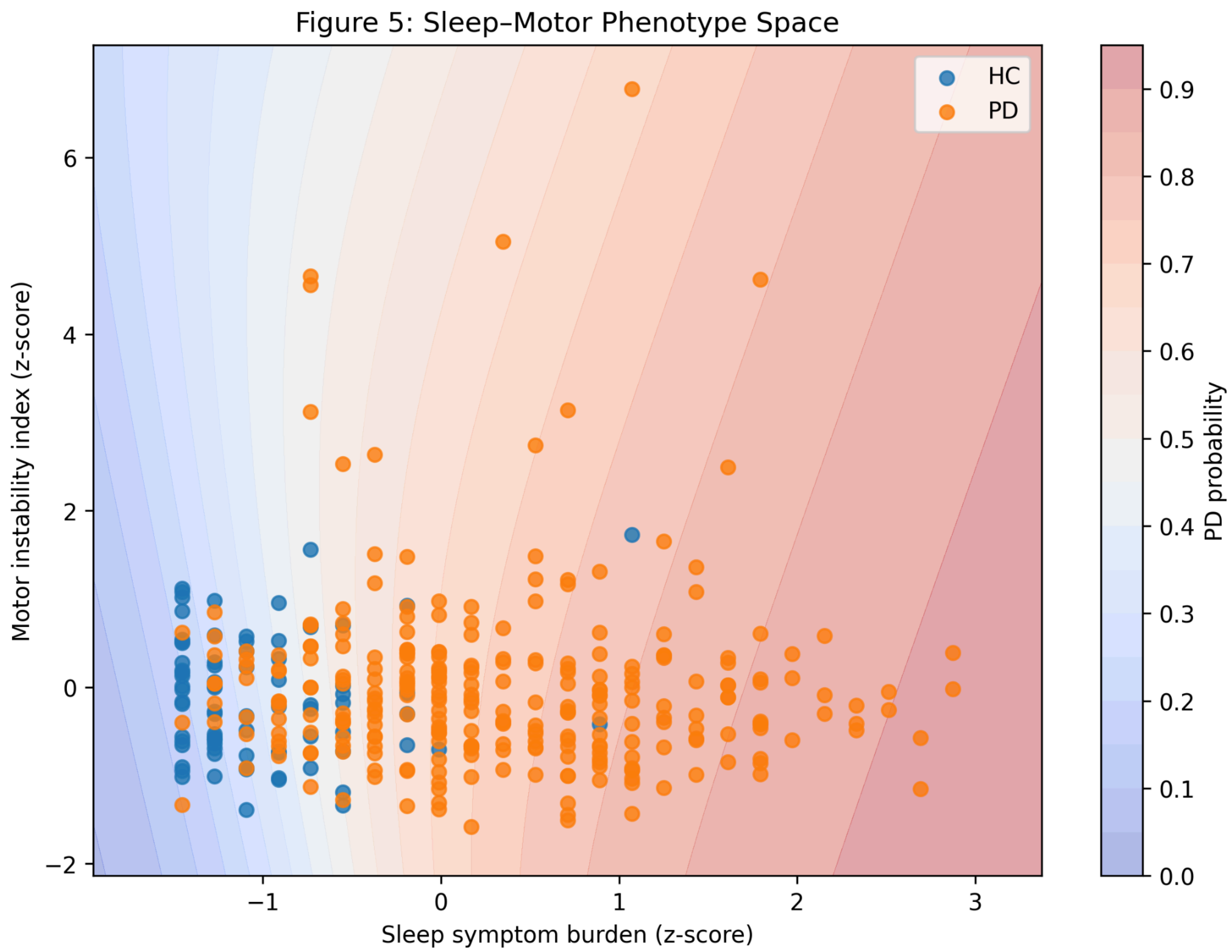

Figure 5. Sleep–motor phenotype space reveals a continuous disease axis separating Parkinson's disease and healthy controls. Subject-level sleep symptom burden and motor instability indices were jointly embedded into a two-dimensional phenotype space. Kernelized decision surfaces derived from a logistic regression model demonstrate a graded probability landscape of Parkinsonian classification across the joint feature space. Healthy controls (HC) and Parkinson's disease (PD) participants occupy partially separable but overlapping regions, indicating that sleep disturbance and motor dysfunction jointly contribute to a continuous, rather than strictly categorical, disease phenotype. This integrated representation highlights a coupled sleep–motor axis underlying Parkinsonian symptom expression and provides a framework for multimodal biomarker modeling in neuropsychiatric and neurodegenerative disorders.

Introduction

Parkinson’s disease is a progressive neurodegenerative disorder characterized by bradykinesia, rigidity, tremor, and postural instability, alongside a broad spectrum of non-motor symptoms including sleep disturbances, mood disorders, and autonomic dysfunction (Kalia & Lang, 2015; Poewe et al., 2017). While motor symptoms have traditionally defined clinical diagnosis, non-motor symptoms are increasingly recognized as critical determinants of disease burden and progression.

Sleep disturbances are among the most common non-motor manifestations in PD, affecting up to 80% of patients (Videnovic & Golombek, 2013). These include insomnia, REM sleep behavior disorder, excessive daytime sleepiness, and fragmented sleep architecture. Emerging evidence suggests that sleep dysfunction may not only co-occur with motor symptoms but may also reflect shared underlying neurodegenerative processes affecting brainstem and basal ganglia circuits (Zhang et al., 2021).

Recent advances in wearable technology have enabled continuous monitoring of motor behavior in naturalistic settings, offering new opportunities for objective quantification of disease severity (Hammerla et al., 2016). Concurrently, computational psychiatry and neurology approaches emphasize the importance of integrating multimodal data sources to construct latent disease phenotypes (Stephan et al., 2016).

However, a key gap remains: the relationship between sleep symptom burden and real-world motor instability has not been systematically quantified using high-resolution wearable data. Here, we propose a computational framework that integrates smartwatch-derived motor features with questionnaire-based sleep assessments to investigate whether sleep dysfunction is coupled with motor impairment in PD.

We hypothesize that (1) PD patients will exhibit increased motor instability compared to healthy controls, (2) sleep symptom burden will be elevated in PD, and (3) sleep burden will be positively associated with motor instability, forming a continuous disease phenotype axis.

Methods

Data Source

We analyzed a multimodal smartwatch-based Parkinson’s disease dataset comprising wrist-worn inertial measurement unit (IMU) recordings and structured symptom questionnaires. The dataset included participants diagnosed with Parkinson’s disease (PD) as well as age-matched healthy control (HC) individuals. Wearable data consisted of tri-axial accelerometer and gyroscope signals collected from both left and right wrists during standardized motor tasks. In parallel, participants completed a structured questionnaire assessing motor and non-motor symptom burden, including sleep-related and neuropsychiatric domains.

Each subject’s data were linked via unique identifiers and organized into synchronized observation files containing metadata, channel labels, and time-aligned sensor recordings.

Signal Preprocessing

Raw smartwatch signals were processed using a standardized preprocessing pipeline to ensure comparability across subjects and recording sessions. Signals were first segmented into predefined motor task epochs (e.g., resting, postural holding, and guided movement tasks). Multi-channel alignment was performed across sensor modalities and wrist locations.

To reduce noise and physiological artifacts, we applied trend removal using an L1-trend filtering approach to suppress low-frequency drift and gravitational acceleration components. This method decomposes each time series into a smooth trend component and a residual dynamic component, allowing isolation of movement-related fluctuations. Following detrending, signals were trimmed to remove initial transient artifacts associated with device vibration and task initiation.

All processed signals were normalized prior to feature extraction to reduce inter-subject variability in recording scale and sensor calibration.

Motor Feature Extraction

Motor behavior was quantified using time-domain and distributional features derived from processed inertial signals. For each subject and sensor channel, we computed:

- **Signal variability:** standard deviation of the time series, capturing overall movement dispersion
- **Signal energy:** root mean square (RMS) amplitude reflecting movement intensity
- **Signal complexity:** entropy of the empirical amplitude distribution, capturing irregularity and unpredictability of motor output
- **Left–right asymmetry index:** absolute difference in aggregated signal magnitude between left and right wrists, quantifying lateral motor imbalance

Features were computed across all relevant channels and then aggregated at the subject level by averaging across tasks and sensor dimensions. The resulting feature set was combined into a composite **motor instability index**, defined as a weighted combination of variability and energy metrics.

Sleep Symptom Quantification

Sleep-related symptom burden was derived from structured questionnaire responses covering multiple domains of non-motor symptomatology. Items included insomnia severity, excessive daytime sleepiness, vivid dreaming, REM sleep behavior–like symptoms, nocturia, and nocturnal motor disturbances.

Responses were encoded numerically and normalized across subjects to ensure comparability across symptom scales. A composite **sleep burden score** was constructed by averaging standardized item responses within the sleep domain. Higher scores reflected greater sleep disturbance severity.

Statistical Analysis

Group-level differences between Parkinson's disease and healthy control participants were assessed using descriptive and inferential statistics applied to both motor and sleep-derived features.

To quantify the relationship between sleep dysfunction and motor impairment, we performed linear regression analysis with motor instability as the dependent variable and sleep burden as the independent variable. Feature values were standardized prior to analysis to ensure comparability of effect sizes.

Additionally, we constructed a multivariate logistic regression model using joint sleep and motor features to estimate disease probability. This model was used to visualize the structure of the combined sleep–motor phenotype space and assess whether the two modalities jointly separate PD from HC participants.

All analyses were performed at the subject level using aggregated feature representations derived from raw time-series and questionnaire data.

Results

Multimodal Study Design and Data Structure

Figure 1 presents the overall study design and computational workflow. The dataset integrates wrist-worn smartwatch recordings and structured questionnaire responses collected from individuals with Parkinson's disease (PD) and age-matched healthy controls (HC). Raw inertial sensor data (accelerometer and gyroscope signals from left and right wrists) were processed through a standardized pipeline including preprocessing, segmentation, and denoising. Parallel questionnaire data capturing sleep and non-motor symptom burden were encoded into structured digital phenotypes. These multimodal inputs were then integrated through feature extraction and statistical modeling to enable joint analysis of motor and sleep-related dysfunction.

Motor Signatures Differentiate Parkinson's Disease from Controls

As shown in Figure 2, Parkinson's disease participants exhibited marked alterations in wearable-derived motor signatures compared to healthy controls. Specifically, PD subjects demonstrated increased signal variability and elevated movement energy, consistent with impaired motor control and increased movement instability. These differences were evident in both raw time-series representations and derived feature distributions, indicating robust separation in motor dynamics between groups.

Sleep Symptom Burden Is Elevated in Parkinson's Disease

Figure 3 demonstrates that sleep-related symptom burden was significantly elevated in PD participants relative to healthy controls. Across multiple domains—including insomnia, excessive daytime sleepiness, nocturnal awakenings, and REM sleep–related disturbances—PD participants exhibited higher symptom severity. These findings highlight the broad involvement of sleep dysfunction as a core non-motor feature of Parkinson's disease.

Wearable-Derived Motor Biomarkers Distinguish Clinical Groups

Figure 4 shows that aggregated motor features derived from smartwatch signals provide clear discrimination between PD and HC participants. Feature-based representations of motor variability and movement energy were consistently elevated in PD, reflecting greater motor instability. These results confirm that wearable-derived digital biomarkers capture clinically meaningful motor abnormalities in naturalistic movement data.

Sleep–Motor Coupling Defines a Continuous Disease Phenotype

Figure 5 demonstrates a significant coupling between sleep symptom burden and motor instability. Joint modeling of these modalities revealed a positive association, whereby increased sleep dysfunction corresponded to greater motor impairment. Rather than forming discrete clusters, participants occupied a continuous phenotype space spanning healthy to Parkinsonian states. Logistic regression over the joint feature space revealed a graded probability surface separating PD from HC, indicating that Parkinson's disease may be better conceptualized as a continuous sleep–motor disorder axis rather than a purely categorical condition.

Conclusion

This study presents a multimodal computational framework that integrates wearable-derived motor signatures with structured sleep-related symptom questionnaires to characterize Parkinson's disease (PD). Using smartwatch-based inertial sensor data and digital symptom phenotyping, we demonstrate that PD is associated with both increased motor instability and elevated sleep symptom burden relative to healthy controls. Importantly, these two domains are not independent; rather, they exhibit a significant positive association, forming a continuous sleep–motor phenotype axis.

This joint structure suggests that Parkinson's disease may be better conceptualized as a distributed, systems-level disorder spanning both motor and non-motor domains rather than as a collection of isolated symptoms. By integrating real-world motor behavior with subjective symptom burden, our framework provides a scalable approach for deriving digital biomarkers that reflect underlying disease severity in a continuous manner.

These findings support the potential of multimodal wearable and digital health data to enable more sensitive and ecologically valid characterization of neurodegenerative disease, with implications for early detection, monitoring, and stratification of Parkinson's disease.

Limitations

Several limitations should be considered when interpreting these findings. First, the study is cross-sectional in nature, limiting the ability to infer temporal or causal relationships between sleep dysfunction and motor impairment. Longitudinal data would be required to determine whether changes in sleep burden precede or track progression of motor symptoms over time.

Second, motor features were derived from a predefined set of wearable sensor recordings during structured tasks, which may not fully capture the complexity of spontaneous real-world movement. Although smartwatch data provide ecologically valid measurements, task-based recordings may introduce context-dependent variability that does not generalize across all daily activities.

Third, sleep symptom burden was assessed using self-reported questionnaire data rather than objective sleep measurements such as polysomnography or actigraphy. While these questionnaires capture clinically relevant symptom domains, they remain subject to recall bias and subjective interpretation.

Fourth, the sample size and dataset composition may limit generalizability. Although healthy controls and Parkinson's disease participants were included, potential heterogeneity within disease subtypes (e.g., tremor-dominant vs. postural instability gait disorder phenotypes) was not explicitly modeled in the current analysis.

Finally, while the observed sleep–motor coupling is statistically robust, the mechanistic basis of this relationship remains speculative. Future work integrating neuroimaging, electrophysiology, or longitudinal wearable monitoring will be necessary to elucidate the underlying neurobiological pathways linking sleep dysfunction and motor impairment in Parkinson's disease.

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